Course notes

Reminders: conditioning and independence

If $P(A) > 0$, $P_A(B) = \frac{P(A \cap B)}{P(A)}$. A tree branch carries a conditional probability; the probability of a path is the product of its branches. For a partition $A_1, \dots, A_r$, $P(B) = \sum_i P(A_i \cap B)$; if all the $P(A_i) > 0$, it can be written $\sum_i P(A_i) P_{A_i}(B)$. Two events are independent when $P(A \cap B) = P(A)P(B)$. For $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. Do not confuse $P_A(B)$ and $P_B(A)$.

Example. $P(A) = 0{,}4$, $P_A(B) = 0{,}7$, $P_{\overline{A}}(B) = 0{,}2$. Then $P(B) = 0{,}4 \times 0{,}7 + 0{,}6 \times 0{,}2 = 0{,}4$ and $P_B(A) = \frac{0{,}28}{0{,}4} = 0{,}7$.

Bernoulli and binomial

A Bernoulli trial has two outcomes, success with probability p and failure with probability $1-p$. The indicator variable B is 1 for a success and 0 otherwise: $E(B)=p$, $V(B)=p(1-p)$. A Bernoulli scheme is a repetition of n identical and independent trials. The number X of successes follows the binomial distribution $\mathcal{B}(n,p)$.

Distribution and parameters

For $k \in \{0,\dots,n\}$,

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}.$$

$$E(X)=np, \qquad V(X)=np(1-p), \qquad \sigma(X)=\sqrt{np(1-p)}.$$

Proof of the distribution. Each path comprising k successes has prob. $p^k(1-p)^{n-k}$. There are $\binom{n}{k}$ choices for the positions of the successes. These paths are mutually exclusive; add their probabilities.

Cumulative events

$$P(X\leq k)=\sum_{j=0}^{k}P(X=j), \qquad P(X\geq k)=1-P(X\leq k-1).$$

$$P(a\leq X\leq b)=P(X\leq b)-P(X\leq a-1).$$

For at least one success, $P(X\geq 1)=1-(1-p)^n$. The complement of "at least k" is "strictly fewer than k".

Example. For $X\sim\mathcal{B}(10,0{,}2)$, $E(X)=2$ and $V(X)=1{,}6$. $P(X=2)=45(0{,}2)^2(0{,}8)^8\simeq 0{,}3020$ and $P(X\geq 1)=1-0{,}8^{10}\simeq 0{,}8926$. The expectation is neither a certain outcome nor necessarily the most likely.

Threshold and interval

To look for the smallest n such that $1-(1-p)^n\geq q$, a loop over n is enough. Using logarithms, we can solve $(1-p)^n\leq 1-q$ then round in the right direction. To find the smallest integer k such that $P(X>k)\leq\alpha$, increase k until obtaining $P(X\leq k)\geq 1-\alpha$, then check the previous index. We can also look for an interval $[a,b]$ containing at least a prescribed probability, without assuming that it is unique.

Independent trials, with more than two outcomes

For independent experiments, the probability of a list of results is the product of the probabilities of its components. The sample space is the Cartesian product of the sample spaces of each experiment. By example, for two independent rolls of a fair die, the sample space contains the 36 pairs (i,j), $1\leq i,j\leq 6$, each with probability $\frac{1}{36}$. The probability that the sum is 5 equals $\frac{4}{36}=\frac{1}{9}$, as the favourable pairs are $(1,4),(2,3),(3,2),(4,1)$. In a Bernoulli scheme, success is coded by 1 and failure by 0: the sample space is $\{0,1\}^n$. Independence concerns the whole sequence; a simple equality between two observed probabilities is not enough to prove it.

Build the model before calculating

13.1 From the table to the tree

First use: S36.

Worked example · Changing the denominator

Out of 1,000 objects, 600 come from workshop A and 400 from B. A produces 12 defective objects and B produces 20. Denoting by D a defect, $P(A \cap D) = 12/1000$, $P_A(D) = 12/600 = 0{,}02$ and $P_D(A) = 12/32 = 0.375$. The same count 12 appears, but the reference sample space changes.

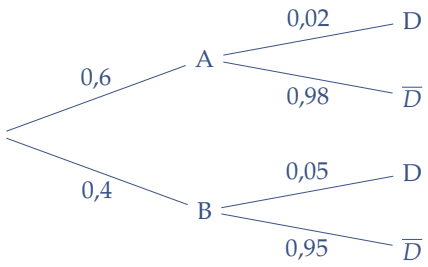


Figure 13.1. Read path A then D: $0.6 \times 0{,}02 = 0.012$, or 12 items out of 1,000.

A1 · Immediate application Complete the table with two rows A/B and two columns D/not D. Calculate $P(D)$ in two ways. Are the defect and the workshop independent?

13.2 Recover a branch and reverse a conditioning

First use: S38.

Property · Partition and the law of total probability

If A and $\overline{A}$ form a partition, $P(D) = P(A)P_A(D) + P(\overline{A})P_{\overline{A}}(D)$, when the conditional probabilities are defined. This result comes from the disjoint union of the two paths leading to D.

Worked example · A missing piece of data

We know $P(A) = 0{,}7$, $P_A(D) = 0{,}1$ and $P(D) = 0{,}13$. Let $x = P_{\overline{A}}(D)$. The equation $0{,}13 = 0{,}7 \times 0{,}1 + 0{,}3x$ gives $x = 0{,}2$. The result belongs to $[0;1]$, which is a necessary check.

A2 · Partial guidance With $P(A) = 0{,}4$, $P_A(D) = 0.2$ and $P(D) = 0.14$, find $P_{\overline{A}}(D)$ then $P_D(A)$.

A3 · Refute "If 90% of patients test positive, then 90% of people who test positive are ill." Build a table for 1,000 people, with 100 ill, 90 positive among the ill and 90 positive among those who are not ill, then calculate the two proportions.

13.3 Four features to check

First use: S39 and S41.

Definition · Bernoulli scheme

We repeat a fixed number n of independent trials, with two outcomes and the same probability of success p. The number X of successes follows the distribution $\mathcal{B}(n;p)$: $P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$ for $0\leq k\leq n$.

Worked example · Counting paths

For four trials with probability 1 /2, getting exactly two successes corresponds to six paths, each with probability 1 /16. Thus $P(X=2)=6/16$. Independence allows us to multiply the probabilities along a path; the binomial coefficient counts the placements of the successes.

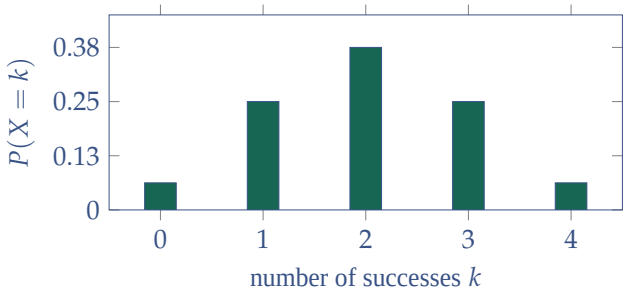


Figure 13.2. Law $\mathcal{B}(4;1/2)$: explain the symmetry and check the sum of the heights.

A4 · Accept or refuse State in which cases the binomial model is justified: ten independent tosses of a coin; ten draws without replacement from an urn containing five red balls and five blue ones ; number of tosses before the first success. Justify each answer.

13.4 Translate words into complete events

First use: S41.

Worked example · A boundary shifted by one unit

For X integer, "strictly more than 4" means $X\geq 5$ and its complement is $X\leq 4$. "At least 4" means $X\geq 4$ and its complement is $X\leq 3$. Finally, $P(2\leq X\leq 5)=P(X\leq 5)-P(X\leq 1)$.

A5 · Immediate application For $X\sim\mathcal{B}(8;0{,}3)$, write in terms of cumulative probabilities the events: at least 3 successes; strictly fewer than 3; between 2 and 5 inclusive. Calculate exactly $P(X=0)$.

13.5 Find a probability rather than a number of trials

First use: S42.

Worked example · Two trials

For two independent trials with probability p, the probability of at least one success is $1-(1-p)^2$. Making it equal to 3/4 amounts to $(1-p)^2 = 1/4$. Since $1-p \geq 0$, we keep $1-p = 1/2$, so p=1/2. The root with the negative sign would be incompatible with p in [0;1].

A6 · Choice of strategy Three trials are independent, with probability of success p. We want the probability of at least two successes to be 0,9. Express this probability in terms of p, prove the uniqueness of a solution in $[0;1]$ and bound it between two consecutive hundredths.

A first step: Count two and three successes separately. For uniqueness, study a function on $[0;1]$.

One minute, spoken aloud. Explain why a probability tree does not imply independence: the probabilities at the second level may depend on the first outcome.

Delayed review · S50 then S77. Solve a threshold $1-(1-p)^n \geq c$ by logarithm, then compare exact probabilities and concentration bounds.

Investigation

Checking components - 40 min.

A component is chosen at random from the output of two workshops. Let A be event "the component comes from the first workshop", $B = \overline{A}$ and D the event "the component is defective". We are given $P(A) = 0{,}6$, $P_A(D) = 0{,}02$ and $P_B(D) = 0{,}05$. Approximate results will be rounded to 10^{-4}.

1. Represent the situation with a weighted tree, then calculate $P(D)$.

2. The chosen component is defective. Calculate the probability that it comes from the first workshop.

3. Twenty components are now chosen independently, each following the previous model. Let X be the number of defective components among the twenty. Justify the distribution of X and specify its parameters.

4. Calculate $P(X=0)$, $P(X \geq 2)$ and $E(X)$.

5. If the twenty components were taken from the same batch, which hypothesis of the model would need to be re-examined? Explain why.